

DPython Business Problem-Solving Practice

1. KPI & Aggregation Analysis

Q1

Management wants a five-year performance summary. Calculate total revenue, total units sold, total purchase cost, total gross profit, and gross margin percentage.

Q2

Which locations generated the highest total revenue during the five-year period?

Q3

Compare the seven product categories based on revenue, units sold, gross profit, and gross margin.

Q4

Which months generated the highest and lowest total revenue?

Q5

Which brands generated the highest total units sold?

Q6

Calculate the average revenue generated per product across the complete dataset.

Q7

Which locations have above-average revenue but below-average gross margin?

Q8

For each package size, calculate revenue, units sold, gross profit, and average selling price.

2. Conditional & Category-Based Analysis

Q9

Classify products into High, Medium, and Low profitability based on gross margin. Compare the revenue generated by each group.

Q10

Identify products that have **high sales volume but below-average gross margin**.

Q11

Identify locations that have **above-average revenue but below-average units sold**.

Q12

Identify products whose selling price is at least 20% higher than their purchase cost.

Q13

Management wants products that satisfy both conditions:

- above-average units sold
- above-average gross profit

Q14

Identify brands that generate high revenue but have relatively weak profitability.

3. Top / Bottom Performer Analysis**Q15**

Identify the top 10 products by total revenue.

Q16

Identify the top five brands by total gross profit.

Q17

For every product category, identify the top three products by revenue.

Q18

For every location, identify the top three products by units sold.

Q19

Identify the bottom 10 products by gross profit.

Q20

Identify the five locations with the lowest revenue.

Q21

Identify the top three brands within every location.

Q22

Which products rank among the top performers based on both revenue and gross profit?

4. Period-over-Period Comparison**Q23**

Calculate month-over-month revenue growth for the entire business.

Identify the months with the largest increase and largest decline.

Q24

Compare each location's monthly revenue with its previous month and identify locations experiencing significant declines.

Q25

Calculate year-over-year revenue growth for each product category.

Q26

Identify product categories whose units sold declined compared with the previous month.

Q27

Identify products whose revenue declined for at least two consecutive months.

Q28

Identify locations that show a consistent improvement in revenue over multiple periods.

Q29

Compare the first year and the final year of the dataset. Which categories improved the most?

Q30

Identify products whose revenue increased while their units sold decreased between two comparable periods. Investigate what may have caused this.

5. Trend & Cumulative Analysis

Q31

Calculate cumulative company revenue month by month from January 2020 to December 2024.

Q32

Calculate cumulative gross profit for every location and compare long-term profitability.

Q33

Which product categories show a sustained upward revenue trend over the five-year period?

Q34

Which product categories show a sustained decline in units sold?

Q35

Determine whether each brand's long-term revenue trend is growing, declining, or relatively stable.

Q36

Identify products that have experienced a major long-term change in sales performance.

Q37

Analyse how the revenue contribution of each product category changed from the beginning to the end of the five-year period.

6. Rolling / Moving-Window Analysis

Q38

Calculate the three-month rolling average of total revenue and use it to understand the underlying sales trend.

Q39

Calculate a six-month rolling average of units sold for each product category.

Which categories show sustained weakness?

Q40

Calculate a three-month rolling revenue measure for every location.

Identify locations whose recent performance is below their normal trend.

Q41

Analyse the rolling gross profit of the major product categories and identify categories whose profitability is weakening.

Q42

Identify products whose recent three-month performance is significantly different from their longer-term performance.

7. First & Latest Event Analysis

For this dataset, an "event" means a **sales observation**, not a customer event.

Q43

For every product, identify its first recorded sales month and latest recorded sales month.

Q44

For each brand, compare revenue in its first recorded month with revenue in its latest recorded month.

Q45

For every location, determine the first and latest month in which sales were recorded.

Q46

For each product category, compare its performance during its first available period with its most recent period.

Q47

Identify products that appeared early in the dataset but have very weak or zero activity in later periods.

8. Previous & Next Event Analysis

Q48

Calculate the change in monthly revenue for every location compared with its previous month.

Q49

For every product, calculate the change in units sold from the previous month.

Identify unusually large changes.

Q50

Identify product-location combinations where sales suddenly increased or decreased compared with the previous month.

Q51

For each brand, calculate month-to-month revenue changes and identify periods of high volatility.

Q52

For every product, compare its current month with its previous month and identify the largest positive and negative movements.

9. Duplicate & Record-Quality Analysis

Q53

Check whether the dataset contains exact duplicate records.

Q54

Investigate whether multiple records exist for the same **Date + Location + Product ID** combination.

Determine whether those duplicates are legitimate or potentially problematic.

Q55

Check whether the same Product ID is associated with different product names, categories, brands, or package sizes.

Q56

Identify products whose selling price changes across different locations or months.

Determine whether this appears to be a legitimate business variation or a data-quality issue.

Q57

Investigate whether the same product appears under inconsistent naming conventions.

Explain how you would clean the data before analysis.

10. Segmentation & Classification Analysis

Q58

Divide locations into High-, Medium-, and Low-revenue segments. Compare the characteristics of each group.

Q59

Divide products into four groups using sales volume and gross margin:

- High volume / High margin
- High volume / Low margin
- Low volume / High margin
- Low volume / Low margin

Identify the products in each group.

Q60

Segment brands according to their contribution to total revenue and identify the strategically important brands.

Q61

Segment products based on their profitability and determine which segments deserve attention from management.

Q62

Segment package sizes based on their sales performance and determine which size has the strongest business contribution.

11. Missing, Inactive & Gap Analysis

Q63

Identify product-location combinations with zero sales in one or more months.

Q64

Identify products that had sales previously but later experienced multiple consecutive zero-sales months.

Q65

Find locations where particular categories became inactive for extended periods.

Q66

Identify product-location combinations with irregular sales activity and determine where further investigation is needed.

Q67

Identify months where one or more product categories had no recorded sales at a particular location.

12. Cohort & Retention Analysis

Your Rice Sales dataset does **not** contain genuine customer-level history, so these should **not** be fabricated using the current dataset.

Q68

Identify each customer's first purchase month and create customer cohorts based on that month.

Q69

For customers acquired in each month, calculate how many returned and purchased again in the following month.

Q70

Calculate 30-day, 60-day, and 90-day customer retention.

Q71

Identify customers who made purchases in multiple months and calculate their repeat-purchase rate.

Q72

Compare high-value repeat customers with one-time customers in terms of revenue contribution.

Q73

Identify month-wise segments whose purchase frequency is increasing or declining over time.

13. Relationship & Cross-Entity Analysis

Q74

Analyse the relationship between **brand and location**. Identify brands whose performance is concentrated in only a few locations.

Q75

Analyse the relationship between **category and package size**. Identify the dominant package size for each category.

Q76

Determine whether high-revenue locations also tend to have high gross margins.

Q77

Determine whether the brands that sell the largest number of units are also the brands generating the largest revenue.

Q78

Analyse the relationship between selling price and units sold. Determine whether higher-priced products generally sell less.

Q79

Identify products that perform strongly in one location but poorly in another.

Q80

Identify categories that are highly dependent on a particular brand or package size.

14. Contribution & Share Analysis

Q81

Calculate each product category's percentage contribution to total company revenue.

Q82

Calculate each brand's percentage contribution to total revenue and total units sold. Compare the two.

Q83

Calculate every location's percentage contribution to company revenue.

Q84

For each month, determine the revenue contribution percentage of every product category.

Q85

Calculate what percentage of total revenue is generated by the top 10 products.

Q86

Calculate the percentage contribution of each location to total gross profit.

Q87

Identify categories whose share of total revenue is increasing over time.

15. Root-Cause & Drill-Down Analysis

This is where I especially want you to challenge yourself.

Do **not** immediately calculate one number and stop.

Q88

Management says:

"Overall revenue has declined in some months. Find out why."

Investigate the issue from:

Company → Location → Category → Brand → Product

Then identify the major contributors to the decline.

Q89

Management says:

"Some locations generate strong revenue but their profit is lower than expected."

Investigate whether the problem is related to:

- product mix
- selling price
- purchase cost
- units sold
- category mix
- brand mix

Q90

Management asks:

"Which products should receive greater business attention?"

Do not simply choose the highest-revenue products.

Use multiple indicators such as:

- revenue
- units sold
- gross profit
- margin
- trend
- location performance

and create a justified shortlist.

Q91

One category has experienced declining unit sales.

Determine whether the decline is:

- company-wide
- location-specific
- brand-specific
- product-specific

Q92

Management says:

"Sales are growing, but we aren't sure whether business health is actually improving."

Analyse:

Revenue + Units Sold + Selling Price + Purchase Cost + Gross Profit + Gross Margin

and determine whether the statement is supported by the data.

Q93

Find a product or location that looks successful according to one KPI but weak according to another KPI.

Explain the conflicting signals.

Q94

Identify the biggest contributor to overall revenue growth during the five-year period.

Break it down from:

Location → Category → Brand → Product

Q95

Perform a complete business performance investigation and prepare a management-style conclusion:

What is performing well?

What is deteriorating?

Where is the biggest opportunity?

Where is the biggest risk?

What should management investigate next?